

Dietary protein intake throughout childhood is associated with the timing of puberty.

Early puberty onset is associated with hormone-related cancers, but whether diet in childhood influences pubertal timing is controversial. We examined the association of protein intake in early and mid-childhood with the ages at take-off of the pubertal growth spurt (ATO), peak height velocity (APHV), and menarche in girls and voice break in boys using data from the longitudinal Dortmund Nutritional and Anthropometric Longitudinally Designed Study. Among participants who provided 3-d weighed dietary records at 12 mo, 18-24 mo, 3-4 y, and 5-6 y, 112 had sufficient anthropometric measurements between 6 and 13 y to allow estimation of ATO. Life-course plots were used to identify critical periods of total, animal, and vegetable protein intake (percentage of total energy intake) for pubertal timing. At these ages, the association between tertiles of protein intake (T1-T3) and the outcomes was investigated using multiple linear regression analysis. A higher total and animal protein intake at 5-6 y was related to an earlier ATO. In the highest tertile of animal protein intake at 5-6 y, ATO occurred 0.6 y earlier than in the lowest [(mean, 95% CI) T1: 9.6, 9.4-9.9 vs. T2: 9.4, 9.1-9.7 vs. T3: 9.0, 8.7-9.3 y; P-trend = 0.003, adjusted for sex, total energy, breast-feeding, birth year, and paternal university degree]. Similar findings were seen for APHV (P-trend = 0.001) and the timing of menarche/voice break (P-trend = 0.02). Conversely, a higher vegetable protein intake at 3-4 and 5-6 y was related to later ATO, APHV, and menarche/voice break (P-trend = 0.02-0.04). These results suggest that animal and vegetable protein intake in mid-childhood might be differentially related to pubertal timing.